

Proposed Framework: Knowledge-Infused Multimodal Scientific QA System

1. Input Layer

The system takes a **scientific research paper** and a **user question** as input.

Inputs include:

- **Text:** Abstract, introduction, methods, results, discussion
- **Tables:** Experimental results, comparisons, numerical values
- **Figures:** Graphs, plots, diagrams
- **Scientific Knowledge Graph (SKG):** Domain concepts, entities, and relations
- **User Query:** Natural language scientific question

2. Multimodal Information Extraction

Each modality is processed independently to preserve its structure.

- **Text Encoder**
 - Extracts semantic representations from paragraphs and sections
 - Identifies key entities (methods, datasets, metrics)
- **Table Encoder**
 - Parses rows, columns, headers
 - Preserves numerical relationships and units
- **Figure Encoder**
 - Extracts visual features from graphs and charts
 - Associates captions and axis labels with visual content
- **Knowledge Graph Encoder**
 - Retrieves relevant scientific concepts and relations
 - Provides background knowledge and constraints

3. Knowledge Infusion & Alignment

This is the **core novelty** of the framework.

- Aligns **text, tables, and figures** with **knowledge graph entities**

- Links experimental results to:
 - Methods
 - Theoretical concepts
 - Prior scientific knowledge
- Resolves ambiguities using structured knowledge
- Reduces hallucination by grounding answers in facts

4. Cross-Modal Reasoning Engine

Performs **joint reasoning** across all modalities.

- Combines:
 - Textual evidence
 - Numerical table data
 - Visual trends from figures
 - Logical constraints from SKG
- Supports:
 - Comparison questions
 - Why/How questions
 - Result interpretation questions

5. Evidence-Grounded Answer Generation

Generates answers that are **transparent and verifiable**.

Each answer includes:

- Final natural language response
- Explicit evidence links:
 - Text paragraph
 - Table cell(s)
 - Figure region
 - Knowledge graph fact

6. Output Layer

The system outputs:

- **Answer**
- **Supporting Evidence**
- **Explanation Path** (how the answer was derived)

This makes the system suitable for:

- Scientific review
- Research validation
- Academic assistance

7. Evaluation & Comparison

The system is evaluated against **ChatGPT (text-only baseline)** on:

- Answer correctness
- Evidence grounding
- Hallucination rate
- Explainability

Flow (One-Line)

Paper + Question → Multimodal Extraction → Knowledge Infusion → Cross-Modal Reasoning → Evidence-Based Answer